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## ***B.Tech. Degree VII Semester Regular/Supplementary Examination November 2024***

### **ME 19-205-0701 REFRIGERATION AND AIR CONDITIONING (2019 Scheme)**

*(Refrigeration data books are permitted for examination)*

Time. 3 Hours

Maximum Marks 60

**Course Outcomes**

On successful completion of the course, the students will be able to:

- CO1: Understand the basics of different refrigeration cycles.  
 CO2: Study the vapour compression and vapour absorption refrigeration cycles-both theoretical and practical cycles, properties of refrigerants and selection criteria for refrigerants.  
 CO3: Understand the different components of refrigeration systems  
 CO4: Get knowledge of psychrometry, air-conditioning systems and will be able to do the cooling load calculation.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

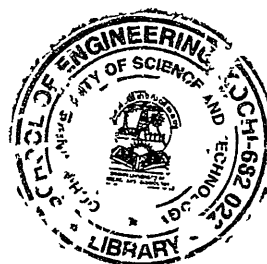
### **PART A (Answer ALL questions)**

|                                                                                                | (8 × 3 = 24) | Marks | BL | CO |
|------------------------------------------------------------------------------------------------|--------------|-------|----|----|
| I. (a) What is meant by coefficient of performance? Explain                                    |              | 3     | L1 | 1  |
| (b) Write a short note on ice refrigeration.                                                   |              | 3     | L2 | 1  |
| (c) How does an absorption refrigeration system differ from a mechanical refrigeration system? |              | 3     | L2 | 2  |
| (d) Write a short note on the selection of a refrigerant for a refrigeration system.           |              | 3     | L2 | 3  |
| (e) What is a flooded evaporator? Explain.                                                     |              | 3     | L2 | 3  |
| (f) What is the effect of clearance volume in a reciprocating compressor?                      |              | 3     |    |    |
| (g) What is the function of a condenser in a refrigeration system?                             |              | 3     | L2 | 3  |
| (h) Classify air conditioning systems as per its use                                           |              | 3     | L1 | 4  |

### **PART B**

|                                                                                                                                                                                                                                       | (4 × 12 = 48) |    |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----|---|---|
| II. (a) A refrigerating system operates on a reversed carnot cycle. The higher temperature of the refrigerant in the system is 35°C and the lower temperature is -15°C. The capacity is to be 12 tonnes Neglect all losses. Determine | 6             | L3 | 1 | 1 |
| (i) Co-efficient of performance.                                                                                                                                                                                                      |               |    |   |   |
| (ii) Heat rejected from the system per hour.                                                                                                                                                                                          |               |    |   |   |
| (iii) Power required.                                                                                                                                                                                                                 |               |    |   |   |
| (b) Draw a neat diagram of simple steam jet refrigeration system and explain its working.                                                                                                                                             | 6             | L2 | 1 | 1 |

OR



(P.T.O)

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- III. (a) One kg of air at a pressure of 1.05 bar and a temperature of 20°C is compressed to 6 bar. It is then cooled to 27°C in the cooler before entering the expansion cylinder. Assuming compression and expansion as isentropic processes, determine  
 (i) Refrigerating effect per kg of air  
 (ii) Theoretical C.O.P  
 Take  $C_p = 1.0 \text{ kJ/kg K}$  and  $\gamma = 1.4$ .
- (b) Explain the working of a simple air cycle cooling system used for aircrafts.

- IV. (a) A F-12 vapour compressor refrigeration system has a condensing temperature of 50°C and evaporating temperature of 0°C. The refrigeration capacity is 7 tonnes. The liquid leaving the condenser is saturated liquid and compression is isentropic. Determine  
 (i) The refrigeration flow rate.  
 (ii) The power required to run the compressor.  
 (iii) The heat rejected in the plant.  
 (iv) C. O. P of the system.

The properties of F-12 are:

| Temp. (°C) | Pressure (bar) | $h_f$ (kJ/kg) | $h_g$ (kJ/kg) | $s_f$ (kJ/kg K) | $s_g$ (kJ/kg K) |
|------------|----------------|---------------|---------------|-----------------|-----------------|
| 50         | 121.99         | 84.868        | 206.298       | 0.3034          | 0.6792          |
| 10         | 3.086          | 36.022        | 187.397       | 0.1418          | 0.6960          |

Take enthalpy at the end of isentropic compression = 210 kJ/kg.

- (b) Draw the T-s and p-h plots of actual vapour compression refrigeration cycle, highlight its differences from a theoretical vapour compression refrigeration cycle.
- OR**
- V. (a) Draw a schematic diagram of an actual vapour absorption refrigeration system and explain its performance.
- (b) Write short notes on  
 (i) Desirable properties of refrigerants.  
 (ii) Need for alternatives to CFC refrigerants.  
 (iii) Primary vs secondary refrigerants
- VI. (a) Differentiate between reciprocating and centrifugal compressors with neat sketches.
- (b) What is heat rejection factor? Discuss its significance?
- OR**
- VII. (a) Explain the working principle of thermostatic expansion valve with the help of a neat diagram.
- (b) Explain the different types of rotary compressors.

(Continued)

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks | BL | CO | PO   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|------|
| VIII. (a) Room air at 20°C DBT and 50% RH is mixed with outdoor air at 40°C DBT and 30% RH in the ratio of 4:1. The mixture is passed through a cooling coil whose temperature is maintained constant at 10°C whose by-pass factor is 0.2. Determine the following:<br>(i) condition of air before entering the coil.<br>(ii) condition of air leaving the coil.<br>(iii) refrigeration load on the cooling coil when 250 m <sup>3</sup> /min of air is supplied to the room.                                                                                                                                                                                       | 7     | L3 | 4  | 1,2, |
| (b) Define the following terms.<br>(i) Saturated air.<br>(ii) Dry bulb temperature.<br>(iii) Dew point temperature.<br>(iv) Relative humidity<br>(v) Specific humidity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5     | L1 | 4  | 1,2  |
| <b>OR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |    |    |      |
| IX. (a) The following data relates to the office air conditioning plant having maximum seating capacity of 25 occupants.<br>Outside design conditions = 34°C DBT, 28°C WBT<br>Inside design conditions = 24°C DBT, 50% RH<br>Solar heat gain = 9120 W<br>Latent heat gain per occupant = 105 W<br>Sensible heat gain per occupant = 90 W<br>Lighting load = 2300 W<br>Sensible heat load from other sources = 11630 W<br>Infiltration load = 14 m <sup>3</sup> /min<br>Assuming 40% fresh air and 60% of recirculated air passing through the evaporator coil and the by-pass factor of 0.15, find the dew point temperature of the coil and capacity of the plant. | 7     | L3 | 4  | 1,2, |
| (b) What are the different factors considered in load estimation sheet for comfort application?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5     | L3 | 4  | 1,2  |

Bloom's Taxonomy Levels

L1 - 52.77%, L2 - 45.98%, L3 - 1.25%.

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